



How to Migrate from SQLite to PostgreSQL in Django

Step 1: Export (Dump) Your Current SQLite Data

Just like before, save all your current data into a JSON file so you can load it into PostgreSQL. Run this in your terminal:

```
python manage.py dumpdata --natural-foreign --natural-primary -e contenttypes -
```

Step 2: Install the PostgreSQL Driver

Django needs a library to talk to PostgreSQL. The easiest one to install (which avoids messy C++ build errors) is the binary version of `psycopg2`. In your terminal, run:

```
pip install psycopg2-binary
```

Step 3: Create the Database in PostgreSQL

Open your PostgreSQL command line (using `psql` or a visual tool like pgAdmin) and create a fresh database:

```
CREATE DATABASE finstella_db;
```

(Note: PostgreSQL automatically handles UTF-8 encoding, so you don't need the extra character set commands like you do in MySQL).

Step 4: Update `settings.py`

Open your Django `settings.py` file, find the `DATABASES` section, and replace it with your new PostgreSQL credentials:

```
# Comment out or delete your old SQLite settings
# DATABASES = { ... }

# Add your new PostgreSQL settings
DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.postgresql',
        'NAME': 'finstella_db',          # The name you used in Step 3
        'USER': 'postgres',             # Your PostgreSQL username (usually 'postg
        'PASSWORD': 'your_password',    # Your PostgreSQL password
        'HOST': 'localhost',            # Or '127.0.0.1'
        'PORT': '5432',                 # Default PostgreSQL port
    }
}
```

Step 5: Run Migrations on the New Database

Tell Django to create all the empty tables in your new PostgreSQL database. Run this in your terminal:

```
python manage.py migrate
```

Step 6: Import Your Data

Finally, push all the data you saved in Step 1 into your fresh PostgreSQL tables:

```
python manage.py loaddata datadump.json
```

You're done! Run `python manage.py runserver`, and your app is now powered by PostgreSQL!